Making Sense of the New Hybrid Access MAC in IEEE 802.11ax/be

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Outline

- Hybrid access MAC protocol
  - Performance analysis - Saturation throughput and access delay
  - Improving performance - Dynamic resource allocation design
- Effectiveness under realistic delay and time-varying channels
- Hybrid access with multi-link operation (MLO)
  - Saturation throughput

Wi-Fi and Its Evolution

Generation	IEEE standard	Adopted	Maximum link rate
WiFi 4	802.11n	2009	6.5 Mbps to 600 Mbps
WiFi 5	802.11ac	2013	6.5 Mbps to 6.9 Gbps
WiFi 6 WiFi 6E	802.11ax	2021	0.4 Mbps to 9.6 Gbps
WiFi 7	802.11be	2024	0.4 Mbps to 23 Gbps
WiFi 8	802.11bn	2028	Up to 100 Gbps

[Source: Wikipedia]

- Wi-Fi is central to providing connectivity
  - In homes, businesses, and even public spaces
- Significant portion of cellular traffic offloaded to Wi-Fi

Wi-Fi and Its Evolution

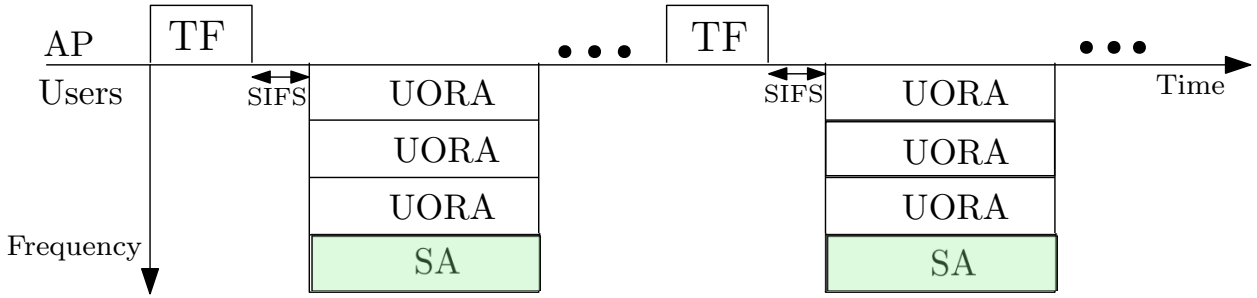
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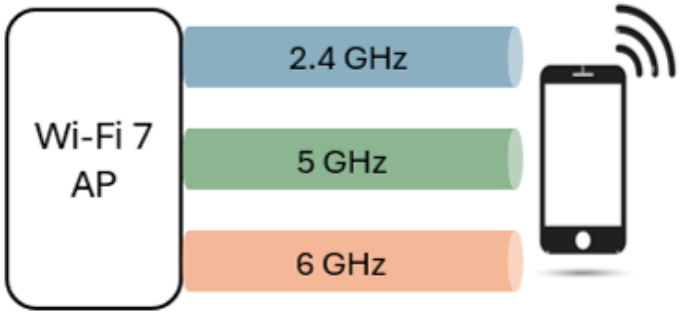
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Wi-Fi: Technological Enhancements

- **IEEE 802.11ax**
  - Physical layer: OFDMA, SU-MIMO and MU-MIMO
  - MAC layer: Hybrid access protocol (UORA + SA)



- **IEEE 802.11be:**
  - Multi-link operation (MLO)

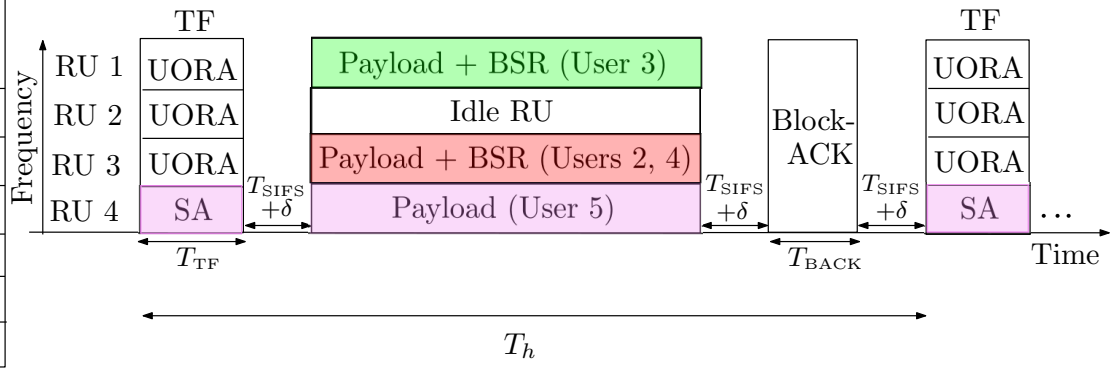


Goal: Understand the optimal use of the key features combined

1. Hybrid Access MAC protocol

- AP allocates RUs for UORA and SA.

User ID	Counter	
	Before TF	After TF
1	7	4
2	1	0
3	2	0
4	3	0
5	–	–
6	–	–



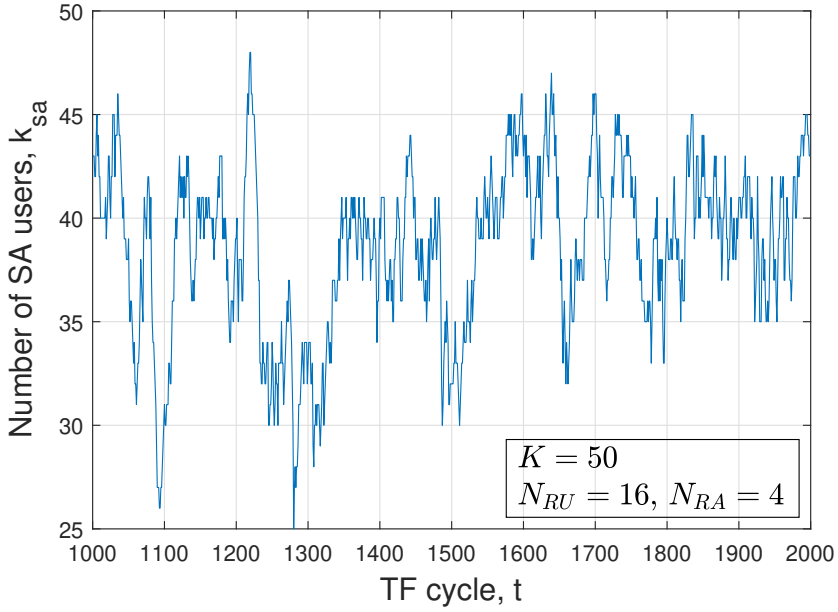
- **Scheduled access:**
  - Cellular-like transmissions.
  - AP schedules SA users.
  - Based on BSRs.
- **Uplink OFDMA random access:**
  - Contention-based.
  - Users randomly choose RUs.
  - Users send BSRs to AP.

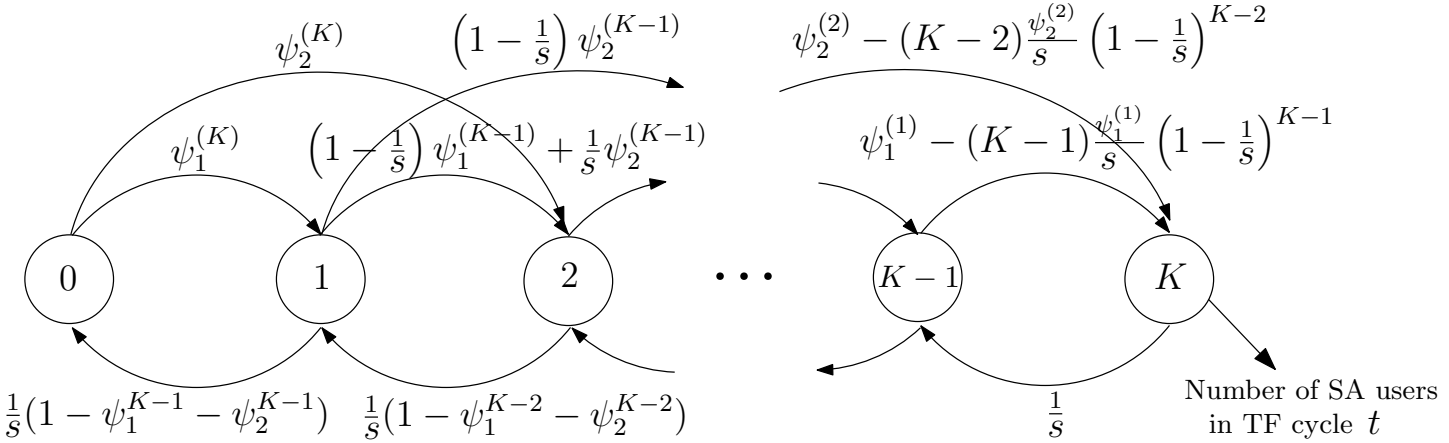
- SA is contention-free and improves throughput.
- UORA enables users to enter SA.

Looking Under the Hood

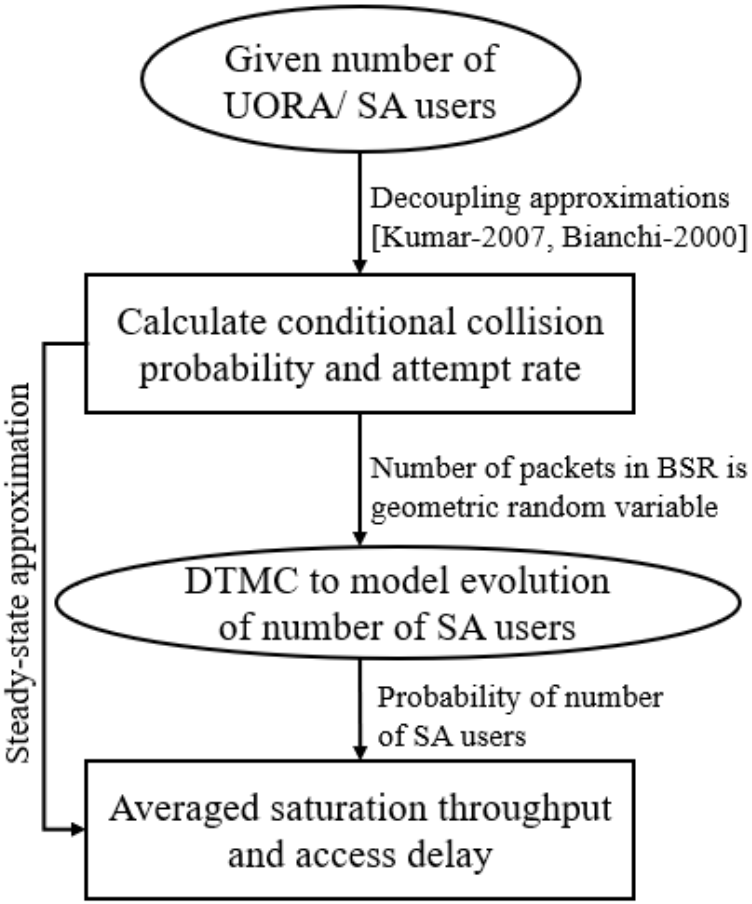
Users move between UORA and SA

- Depends on RU allocation, MAC contention, and time user stays in SA.
- Modeled with a DTMC.



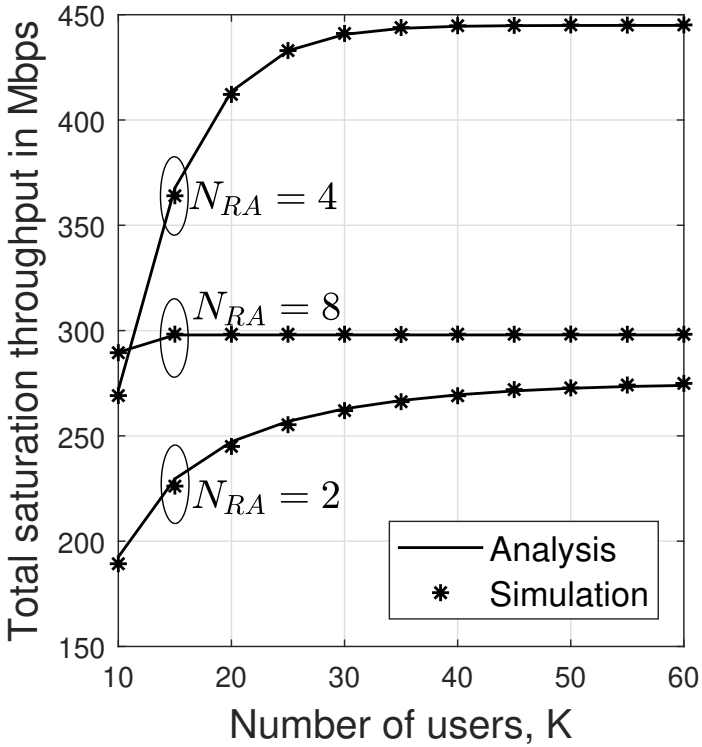


Our Analytical Approach and Goals

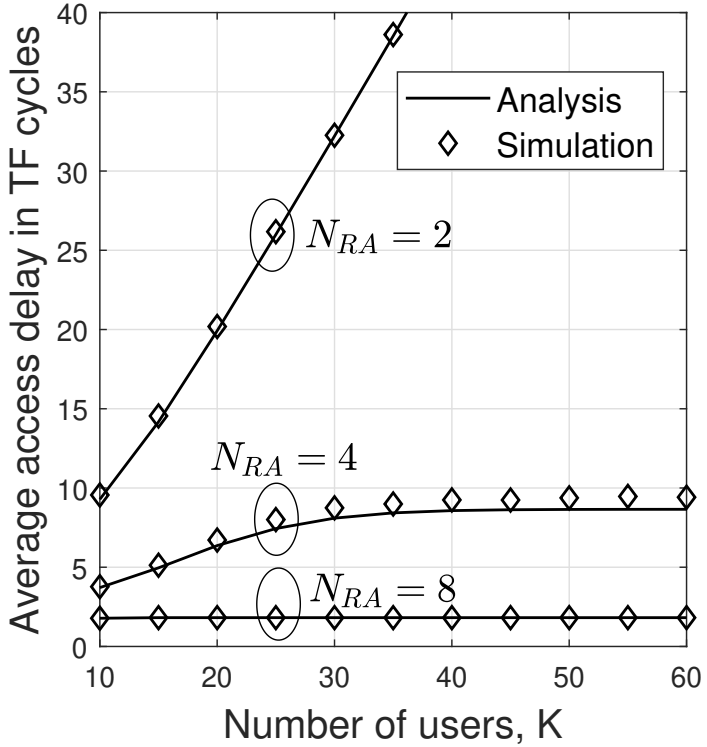


- **Performance Analysis:**
  - Saturation throughput
  - Average access delay
- **Optimal RU allocation:**
  - Trade-off between throughput and access delay.
  - Need RU allocation to balance throughput and access delay.

Saturation Throughput - Access Delay Trade-off



(a) Saturation throughput

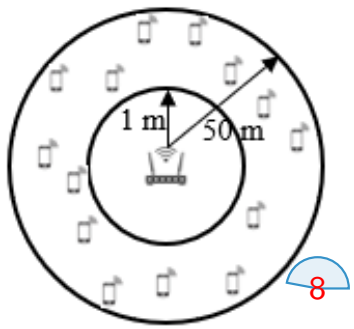


(b) Average access delay

(BW = 160 MHz, $N_{RU} = 16$, $T_P = 5$ ms, and $s = 10$.)

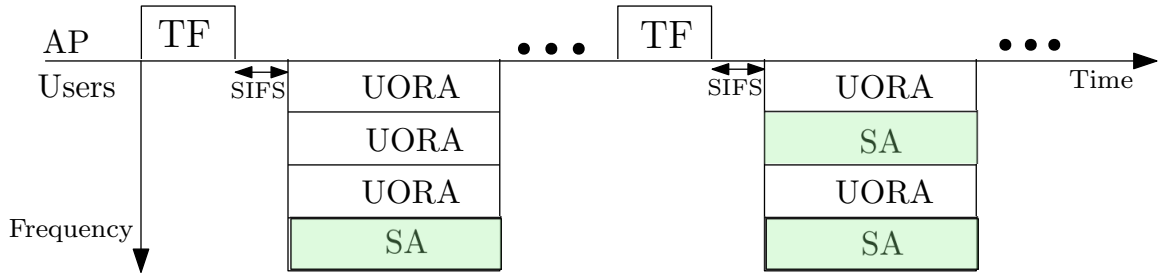
- Trade-off exists between throughput and access delay.

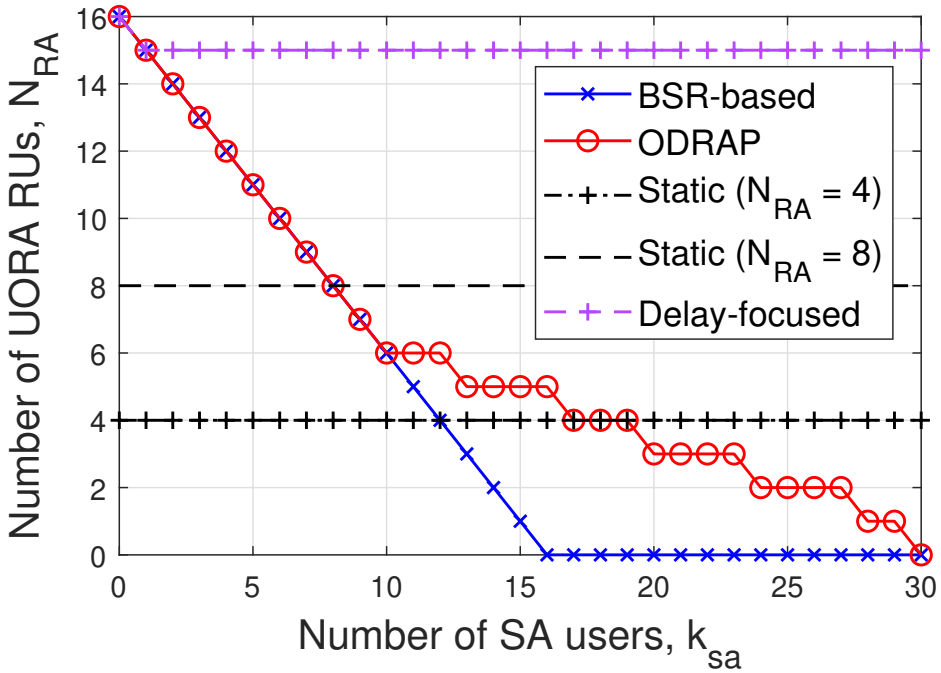
  [S. Arthi and N. B. Mehta, IEEE Globecom 2024]



Optimal RU Allocation Policy

- Dynamic RU allocation - RU allocation depends on number of SA users.



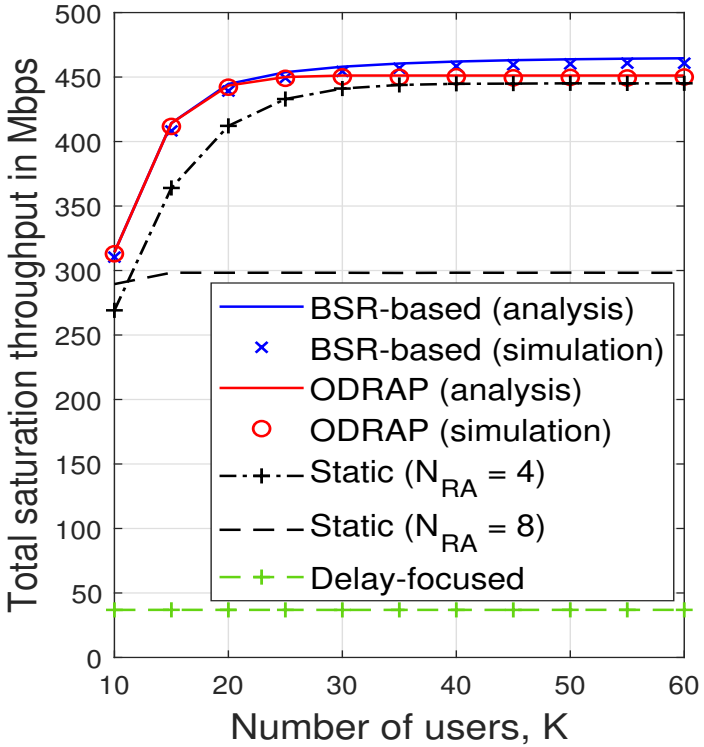


$K = 30$ **and** $N_{RU} = 16$

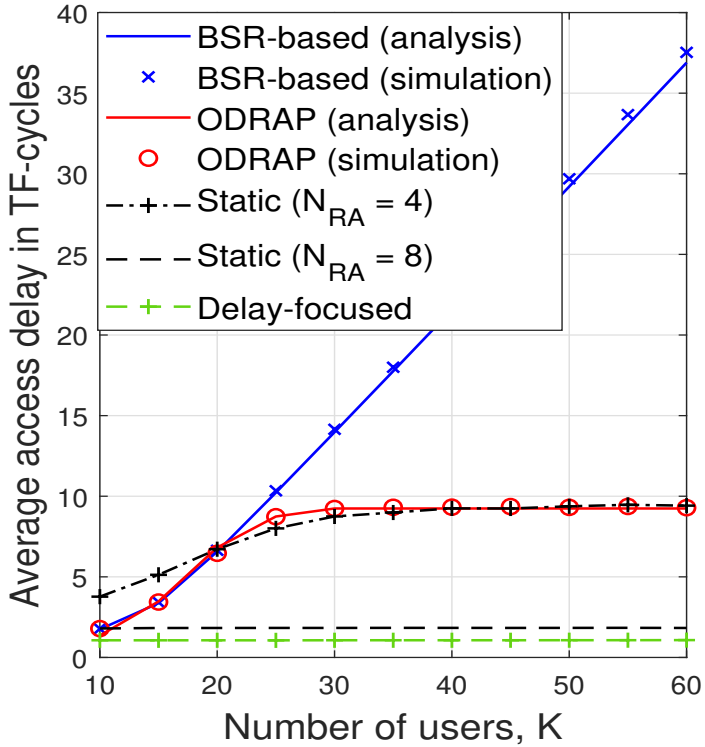
- Based on the DTMC model.
- MDP formulation to find the optimal RU allocation policy.
- Optimally balances throughput and delay.

[Submitted to IEEE TMC]

Optimal Policy: Saturation Throughput and Access Delay



(a) Total saturation throughput



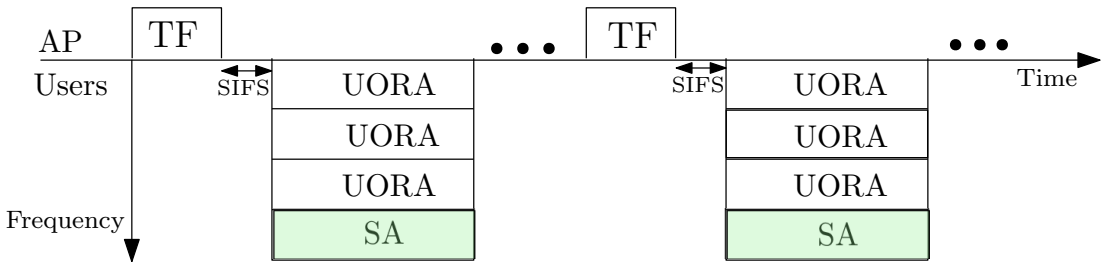
(b) Average access delay

(BW = 160 MHz, $N_{RU} = 16$, $T_P = 5$ ms, and $s = 10$.)

- Throughput of optimal policy is (marginally) lower than BSR-policy.
- But much lower average access delay for large K than BSR-policy.

2. CSI-based MCS Selection in Scheduled Access

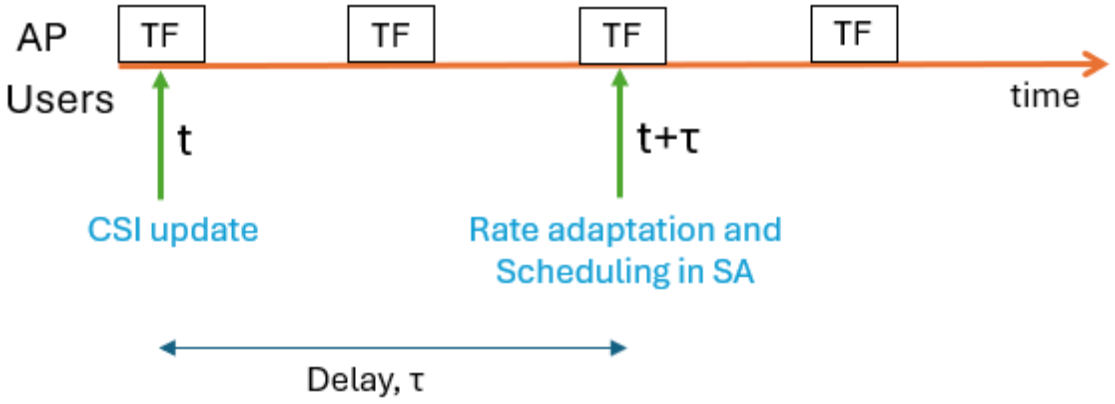
- AP specifies MCS for each of the scheduled users in TF.
  - AP requires the CSI of each SA user in the uplink.

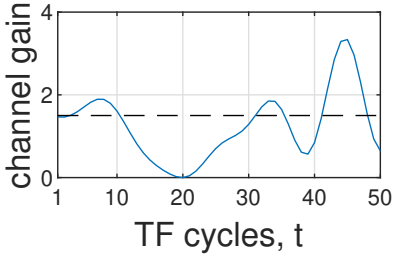


- CSI acquisition approaches in 802.11ax:
  - Typically done through the MAC layer (Minstrel).
  - Channel sounding procedure.
  - CSI updates through uplink preambles.

- **Goal is to characterize the impact of channel variations.**

Impact of Channel Variations



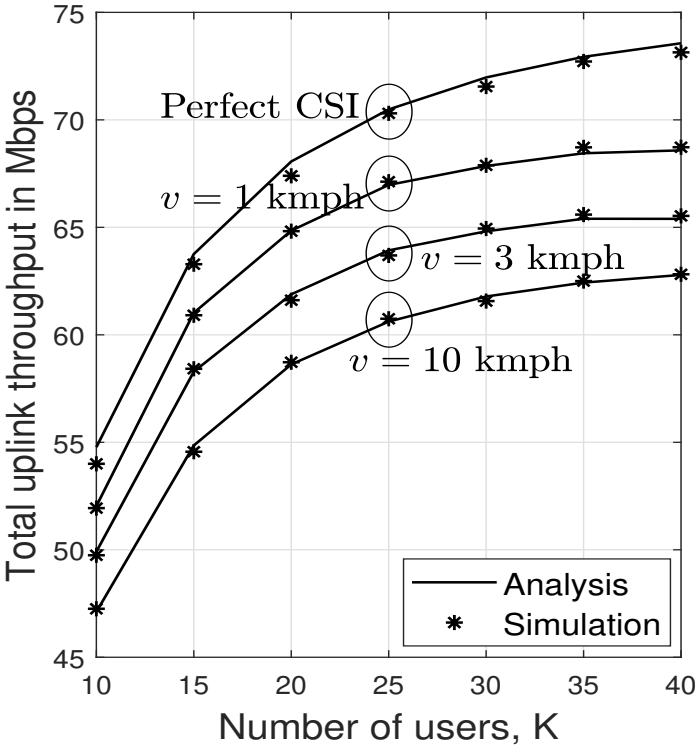


- Correlation coefficients for various CSI delays and speeds ($f_c = 5$ GHz):

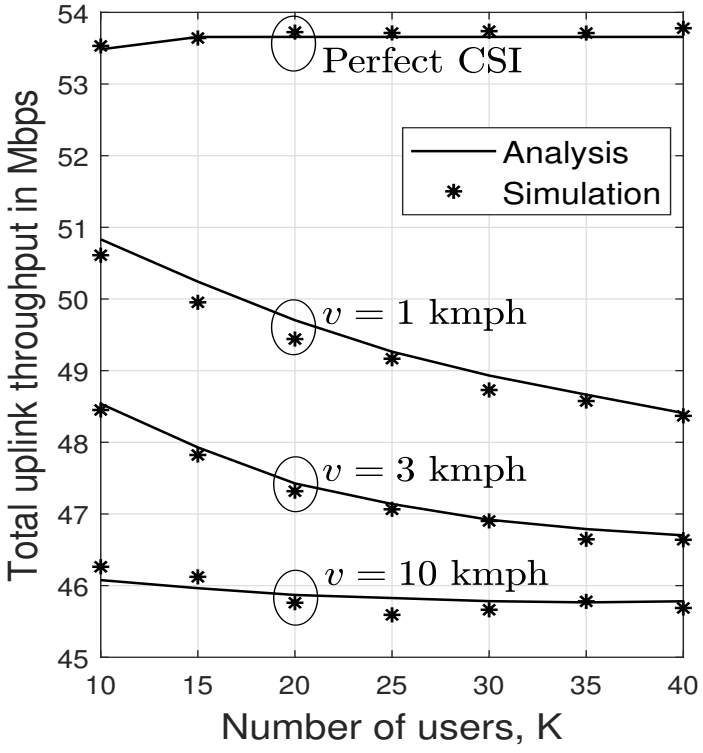
Speed \ CSI delay	3.6 ms	7.2 ms	10.8 ms	14.4 ms	18.0 ms
1 kmph	0.99	0.97	0.94	0.89	0.84
3 kmph	0.94	0.77	0.55	0.32	0.14
10 kmph	0.47	0.00	0.14	0.07	0.01

- Outdated CSI can underestimate/overestimate the channel!
- CSI delays are random.

Impact on Total Throughput



(c) 7 SA RUs

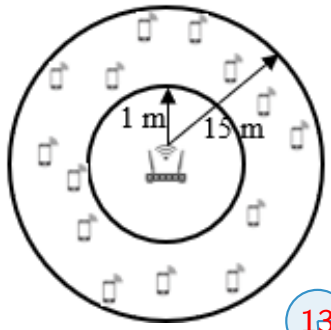


(d) 5 SA RUs

(BW $= 20$ MHz, $N_{RU} = 9$, $T_P = 3$ ms, and $s = 10$.)

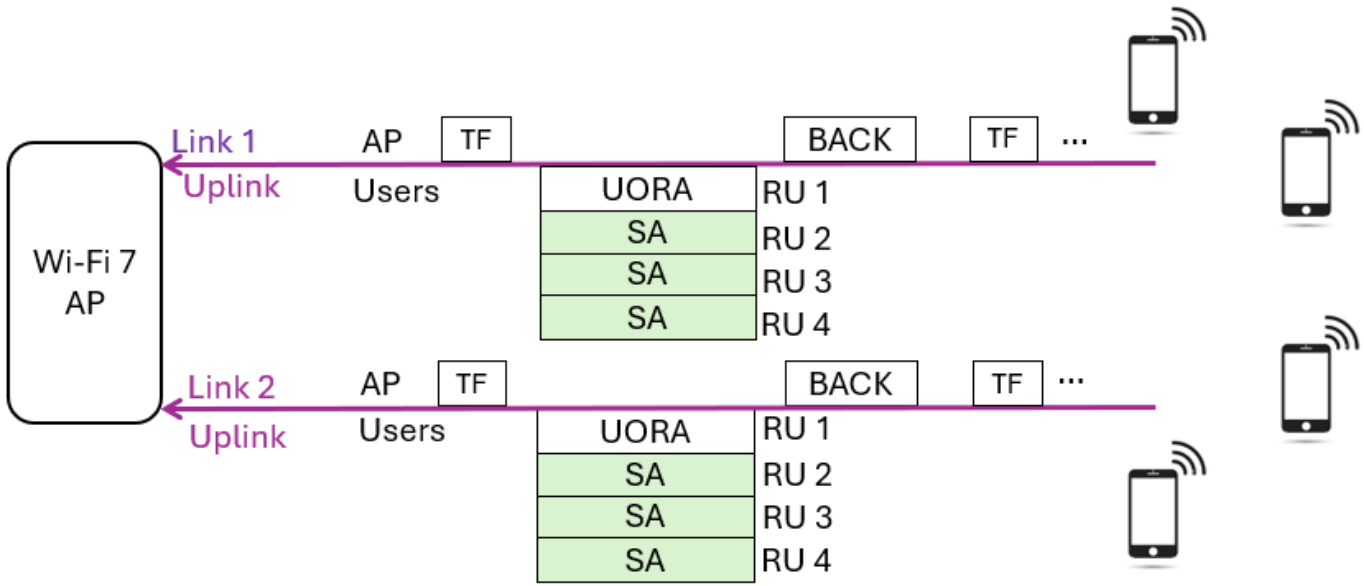
- **Outdated CSI significantly impacts throughput.**

[Submitted to IEEE Globecom 2025]



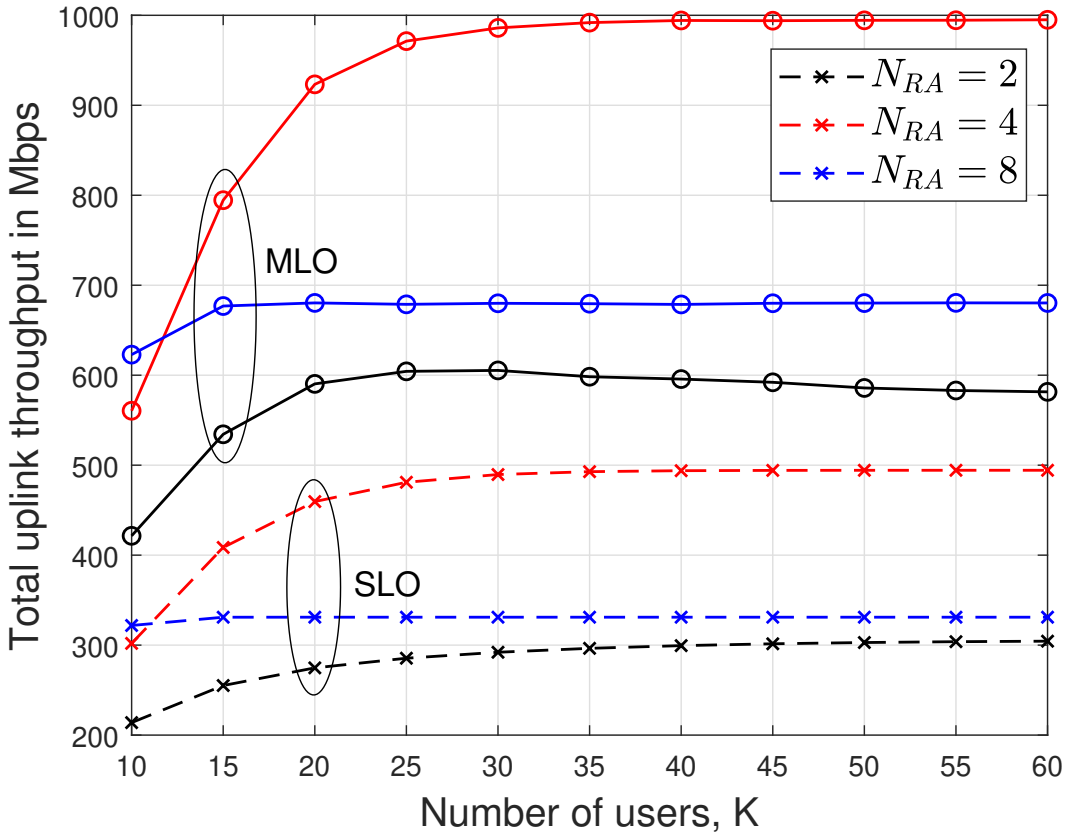
3. Hybrid Access + Multi-link operation

1. Hybrid access uplink + MLO (asynchronous mode)
   - AP sends periodic TFs.
   - All users support MLO and hybrid access.



2. Hybrid access uplink + MLO + Legacy users
   - AP and legacy users contend with conventional DCF.
   - AP sends an TF after every success.

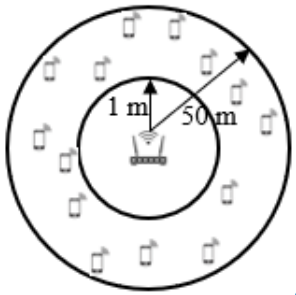
Total Throughputs for Hybrid Access + MLO



[Preliminary results]

(BW = 160 MHz, $N_{RU} = 16$, $T_P = 5$ ms, and $s = 10$.)

- Total throughputs are doubled for MLO.



Summary

- **Analyzed throughput and access delay of hybrid access.**
  - UORA and SA throughputs depend on RU allocation.
  - Our policy optimally balances throughput and delay.
- **Impact of CSI delays in time-varying channels.**
  - Significant channel variations even at low speed.
  - Throughput degradation depends on RU allocation and user speed.
- **Performance of MLO + hybrid access MAC protocol.**
  - MLO can double hybrid-access throughput.

Thank you... Questions?

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